We Don't Talk About The Integers

Kurt Gödel's own showstopper

Looking for structure is one of the most natural tasks for human beings. Watching a landscape and identifying patterns; reading poems and guessing where the beat lies; staring at paintings and figuring out the quiet refrain of ideas, inspirations, stories being told. Structure guides us and reassures us in what is perhaps an extraordinary expression of anthropocentrism, as if the universe arranged itself in recognisable ways just for us to enjoy. While hoping to find patterns in the wild world around us might come off as slightly too optimistic (or even arrogant), for mathematicians this is often the purpose of their work, the end goal of their career: to build structure out of chaos, identify threads of meaning in seemingly abstract objects, single out reason, even when hidden behind a game of smoke and mirrors. Even when they come from far off regions of the mathematical landscape, mathematics often presents itself to mathematicians as a complex puzzle to solve, an enigma to decode. Behind all the complex computations, the red herrings and the chaos, stands some painting of order, hard and confusing and abstract as it may be.

Logicians are no strangers to this phenomenon. For them, mathematics is but a language, to be understood through the lenses of mathematics itself; while this might seem strange at first sight, it is not any stranger than linguists analyzing English through the lenses of English itself. After all, the bread and butter of mathematical research is proofs, which can be modelled to be schematic arguments, sequences of deductions from axioms. When we say that a theorem is true, the logicians argue, what we are really saying is that there is a finite sequence of formal rules that, applied to certain statements which we assume to be true, produces our theorem. It is not much different from arguing about which moves are allowed in a certain game; if you have ever played Dungeons and Dragons, you will have realized that rulebooks don't cover *all possibilities*, and some is left to the players' deduction. Starting from the rules that are written down (the *axioms*), those who approach the game can try and argue that certain moves (the *theorems*) are possible and allowed.

What does linguistic – and, by extension, logic – have to do with this? Any language needs an alphabet, a set of symbols with which we can write down our phrases and statements. If we want to talk about apples, we better have the letters necessary to build the word *apple* in our alphabet. Similarly, if we want to talk about a certain mathematical object, say for example a *ring*, then we need to have symbols to talk about its operations and neutral elements: a symbol + for addition, a symbol $\times$ for multiplication, a symbol 0 for the additive identity, a symbol 1 for the multiplicative identity. To express mathematical statements, we will need to expand our alphabet a little bit. In the most common version of mathematical logic, *first-order logic*, alphabets are assumed to also contain variables (x, y, z...), connectives ($\wedge$, which represents *and*; $\vee$, which represents *or*; $\rightarrow$, which represents *then*; $\neg$, which represents *not*), and quantifiers ($\forall$, which represents *for all*; $\exists$, which represents *there exists*) and a symbol for equality, =. We now have a rather large alphabet; let's try to express some mathematical fact. For example, suppose we want to state *the addition + is a commutative operation*. What this means is that whenever I pick two elements a and b, $a + b$ is the same as $b + a$. In first-order logic, this looks like this:

$$\forall x \forall y (x + y = y + x).$$

First-order logic, together with the language for rings that we have chosen above, allows us to express many interesting properties of rings (though not all). Certain properties will be true in certain rings, and not in others: for example, the operation $\times$ is going to be commutative in the integers $\mathbb{Z}$, but not in the ring of square matrices $M_n(\mathbb{R})$. One can group all statements that are true in a certain ring R in the *theory* of that ring $\mathrm{Th}(R)$. For example, then, the sentence $\forall x \forall y (x + y = y + x)$ is an element of the set $\mathrm{Th}(\mathbb{Z})$, but not of the set $\mathrm{Th}(M_n(\mathbb{R}))$.

While first-order statements can capture many of the important facts about a mathematical object, in this case a ring, they do not typically capture *everything*; in other words, one does not usually have that rings with the same theory are isomorphic. When this happens, it is often because a deeper reason lies behind: once again, some structure to be uncovered, some engine quietly humming in the background. Logicians who deal with these matters are typically labelled as *model theorists*; from this brief reflection emerges one of the fundamental question that a model theorist can ask themselves: how much does $\mathrm{Th}(R)$ know about R?

Enter Shelah.

Saharon Shelah's proposal may seem counterintuitive at first. He poses that there is a fundamental reason why theories don't remember much about mathematical objects, in other words when there are many non-isomorphic objects with the same theory. This loss of information, this emergence of chaos, is produced from certain

patterns, which like Patient Zeros infect the whole object with irredeemable opaqueness. Finding out some of these criminal patterns among the subsets of a certain object that can be described by first-order formulae is sign that the object itself might be untouchable. Objects where these patterns are omitted should be tame, analysable with our toolbox, safe to approach. He isolated a whole hierarchy of "dividing lines" that discern wild objects from tame objects on the basis of the omission of certain patterns. Structure is achieved by omission, removing what might seem like helpful structure – linear orders, trees, certain kinds of graphs, – but is instead a mole, responsible for devastating chaos.

This tentative classification of mathematical objects (not just rings, but also groups, graphs – changing the alphabet allows us to deal with different realms) hits a definite halt when one particular character enters the stage. Long-time fans of true crime know that, whenever some deeply unsettling truth about somebody is unearthed, there will always be some neighbour commenting "Oh, but he was so kind, so polite." No matter if the person in question has committed several gruesome murders, there is a very high chance that those who knew him on a daily basis will say that they would have never expected it. While the community might disagree on which objects are easy, accessible, and which ones aren't, there is a structure so common, polite and kind that nobody would expect it to pull any form of gruesome crime: the integers. After all, 'God created the Integers,' wrote once Kronecker. Model theorists, much like the suspicious officer in a true crime story, are able to see beyond the cute face of this seemingly harmless mathematical structure. They know which sins they are guilty of, even when the rest of the world believes them to be saints, granting miracles to those who praise them. To understand why, we have to go back some 40 years before Shelah's groundbreaking work.

The sound of falling sand

The breaking point of the idilliac picture of the ring of integers and its simple orations is a procedure known as Gödelization. The representation was introduced — as the name suggests — by Kurt Gödel, in an effort to prove his famous *incompleteness* theorems. He sought to argue that certain systems of axioms, as long as they were complex enough, would not be able to prove or disprove all possible statements; what was called being incomplete. Before his work, some believed that it was possible to isolate some axioms and then derive the whole of mathematics from them through formal operations and deductions; this was how, for example, theorems in Euclidean geometry were proved. To some mathematicians, notably Hilbert, it seemed reasonable that the whole of mathematics behaved like Euclidean geometry. They believed that through enough sweat and tears, the community would be able to come up with some basic axioms and then derive all known theorems of mathematics, and all future ones. Proofs would then be a matter of cleverly manipulating symbols and formal systems.

Gödel — and others — disagreed. In the 1930s, he proved that, as long as a formal system (that was *effective*, i.e. could be listed by an algorithm) was able to perform the arithmetic of the integers, in other words had its full ring structure, then it would not be able to prove or disprove all possible statements about them. To do so, he created a translation procedure that transformed first-order statements (like $2 = 2$, or $n + 1 = m + 1 \to n = m$) into (very big) natural numbers. Axioms and proof were encoded into integers and their properties; the provability of a statement, then, was an arithmetical fact about the so-called 'Gödel number' of that statement.

For example, the first-order statement $n + 1 = m + 1 \to n = m$ that I wrote above is encoded with the natural number

$$\begin{aligned} 2 \times 14348907 \times 48828125 \times \\ 232630513987207 \times 4177248169415651 \times \\ 1792160394037 \times 2390724356851513248471 53 \\ \times 322687697779 \times 504036361936467383 \times 29 \\ \times 25408476896404831 \times 45648794082603515540414691 7 \times \\ 550329031716248441 \times \\ 10861771343660416614908294685907 \times 1119130473102767. \end{aligned}$$

This translation procedure allows us to encode entire formal systems inside the integers. Using the ring structure of $\mathbb{Z}$, are able to encode — through very complicated and certainly very long arithmetical sentences — patterns of almost arbitrary difficulty.

There is a fine line between *complex enough to be interesting* and *too complex to be dealt with*. Shelah's framework falls apart in the integers. They sport an impossible, almost supernatural level of expressitivity; one can certainly expect that any conceivable pattern can be encoded or reconstructed inside the ring of integers. Their arithmetic is far too intricate to be decoded through formal properties, making them whatever the opposite of a Holy Grail is for

model theorists. They are a cancer that spreads fast, a toxic creature that poisons everything it touches, making it super tricky to tackle with the available methods.

To many mathematicians, they are the simplest object we can think of; they are, after all, not too far from just putting one pebble after the other, recursively — counting like children count, with our hands. They provide the base to many of our grandiose castles of ideas and theories, the foundations to many explorations into unknown mathematical lands. To model theorists, however, they reveal a different face. A twisted expression, a creepy smile. The source of all darkness, the original Pandora Box. We don't talk about them, hoping that in our cautious adventures we will never find ourselves alone with the ring of integers in a dim-lit alley.